

1. (a) $\frac{\binom{25}{7}\binom{24}{0}}{\binom{49}{7}}$. No R: `dhyper(7, 25, 24, 7)`
 (b) $\frac{\binom{9}{3}\binom{40}{4}}{\binom{49}{7}}$. No R: `dhyper(3, 9, 40, 7)`
 (c) $\sum_{k=5}^7 \frac{\binom{9}{k}\binom{40}{7-k}}{\binom{49}{7}}$. No R: `1-phyper(4, 9, 40, 7)` ou `sum(dhyper(5:7, 9,40,7))`
 2. 0; $\frac{\binom{8}{6}\binom{5}{0}}{\binom{13}{6}}$. No R: `dhyper(6,5,8,6); dhyper(6, 8, 5, 6)`
 3. (a) e^{-3} No R: `dpois(0,3);` (b) $\sum_{k=0}^3 \frac{3^k}{k!} e^{-3}$ No R: `ppois(3,3);` (c) $\sum_{k=5}^{+\infty} \frac{3^k}{k!} e^{-3}$ No R: `1-ppois(4,3)`
 (d) $\frac{\sum_{k=5}^{10} \frac{3^k}{k!} e^{-3}}{\sum_{k=5}^{+\infty} \frac{3^k}{k!} e^{-3}}$ No R: `(ppois(10,3) - ppois(4,3)) / (1-ppois(4,3));` (e) $K = 4$
 4. (a) 0.0988; (b) 0.0141; (c) $f(k) = \begin{cases} e^{-0.6p} \frac{(0.6p)^k}{k!} & \text{se } k \in \mathbb{N}_0 \\ 0 & \text{se } c.c. \end{cases}$ (d) $Poisson(\lambda p)$
- Sug.: Em (c) e (d), use o T.P.T. com a partição $(X = n)_{n \in \mathbb{N}_0}$.
5. (a) 0.21; 0.0189. No R: `dgeom(1,0.7); dgeom(9,0.7)/(1-pgeom(5,0.7))` ou `dgeom(3,0.7)`.
 (b) 0.0002. No R: `1-pgeom(6,0.7)`

1. (a) $k = \frac{1}{2}, F(c) = \begin{cases} \frac{1}{2}e^c & \text{se } c < 0 \\ 1 - \frac{1}{2}e^{-c} & \text{se } c \geq 0 \end{cases}$
 (b) $\frac{1}{2}; \frac{1}{2}; 0; 0; \frac{1}{2}(1 - e^{-1}); \frac{1}{2}(1 - e^{-1}); \frac{1}{2}(1 - e^{-1}); \frac{1}{2}(1 - e^{-1}); 1 - e^{-1}$ (c) $Exp(1)$
2. (a) $F(c) = \begin{cases} 0 & \text{se } c < -1 \\ c + \frac{1}{2}c^2 + \frac{1}{2} & \text{se } -1 \leq c < 0 \\ c - \frac{1}{2}c^2 + \frac{1}{2} & \text{se } 0 \leq c < 1 \\ 1 & \text{se } c \geq 1 \end{cases}$ (b) $0; \frac{7}{8}; \frac{3}{8}; \frac{1}{8}; \frac{5}{9}$
3. (a) $F(c) = \begin{cases} 0 & \text{se } c < 0 \\ 1 - e^{-\lambda c} & \text{se } c \geq 0 \end{cases}$; $P(T > c) = \begin{cases} 1 & \text{se } c < 0 \\ e^{-\lambda c} & \text{se } c \geq 0 \end{cases}$ (b) — (c) $\frac{3e^{-4}}{3e^{-4} + e^{-2}}$
4. $Y \sim Exp(1)$
5. (a) $a = 0; b = 1; k = 2; f(x) = \begin{cases} 0 & \text{se } x \leq 0 \vee x \geq 2 \\ 1 - \frac{x}{2} & \text{se } 0 < x < 2 \end{cases}$ (b) $\frac{1}{16}$ (c) 0.0344 (d) 20kg
6. (a) 0.4718; $\frac{1}{2}$; 0.0562; 0.1147 (b) 0.1442
7. —
8. (a) 0.6827 (b) 0.9545 (c) 0.9973

9. 0.0455

10. (d)

11. (a) 0.25 (b) 0.25 (c) 0.6 (d) 359ml

12. (a) $F(c) = \begin{cases} 0 & se \quad c < 2 \\ \frac{c-2}{10} & se \quad 2 \leq c \leq 12 \\ 1 & se \quad c > 12 \end{cases}$ (b) $U([2, 12])$ (c) 0.6; 0.1209

13. (a) 0.3935 (b) 0.3679 (c) 0.3935

14. (a) — (b) $H_1(t) = \begin{cases} 0 & se \quad t < -1 \\ 1 & se \quad t \geq -1 \end{cases}$ e $H_2(t) = \begin{cases} 0 & se \quad t < \frac{1}{2} \\ \frac{2t-1}{3} & se \quad \frac{1}{2} \leq t \leq 2 \\ 1 & se \quad t > 2 \end{cases}$

15. $P(Y = 0) = 1 - e^{-\lambda a}$, $F_Y(c) = \begin{cases} 0 & se \quad c < 0 \\ 1 - e^{-\lambda(c+a)} & se \quad c \geq 0 \end{cases}$

Soluções da Folha Prática 7

1. —

2. (a)

	$E[X]$	$Var[X]$	σ_X	$\chi_{0.25}$	$\chi_{0.5}$	$\chi_{0.75}$
5. (a)	$\frac{70}{36}$	$\frac{2660}{1296}$	$\sqrt{\frac{2660}{1296}}$	1	2	3
5. (b)	$\frac{161}{36}$	$\frac{2555}{1296}$	$\sqrt{\frac{2555}{1296}}$	3	5	6
5. (c)	$\frac{91}{36}$	$\frac{2555}{1296}$	$\sqrt{\frac{2555}{1296}}$	1	2	3
5. (d)	1	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	0	1	1
5. (e)	1	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	0	1	1
5. (f)	7	$\frac{210}{36}$	$\sqrt{\frac{210}{36}}$	5	7	9
9	$\frac{3}{4}$	$\frac{11}{16}$	$\sqrt{\frac{11}{16}}$	0	0	1

(b)

	$E[X]$	$Var[X]$	σ_X	$\chi_{0.25}$	$\chi_{0.5}$	$\chi_{0.75}$
1.	0	2	$\sqrt{2}$	$-\log(2)$	0	$\log(2)$
2.	0	$\frac{1}{6}$	$\frac{1}{\sqrt{6}}$	$-1 + \frac{\sqrt{2}}{2}$	0	$1 - \frac{\sqrt{2}}{2}$
5.	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{\sqrt{2}}{3}$	$2 - \sqrt{3}$	$2 - \sqrt{2}$	1

3. $E[Y]$ existe e $E[Y] = \frac{1-e^{-\lambda}}{\lambda}$

4. (a) $\frac{1}{2}$ (b) $Y \sim U([0, 1])$, $E[Y] = \frac{1}{2}$ e $Var[Y] = \frac{1}{12}$

5. (a) $F(c) = \begin{cases} 0 & se \quad c < 0 \\ 4c^3 - 3c^4 & se \quad 0 \leq c \leq 1 \\ 1 & se \quad c > 1 \end{cases}$; $E[X] = \frac{3}{5}$ e $Var[X] = \frac{1}{25}$

(b) $E[L] = \frac{9v_1+39v_2+33v_3}{81} - k$; $\sigma_L = \sqrt{\frac{9v_1^2+39v_2^2+33v_3^2}{81} - \left(\frac{9v_1+39v_2+33v_3}{81}\right)^2}$

6. —

7. —